Skills Summary

Building a Coherent Proof from the Givens

Use this reference while completing your worksheet or when studying.

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1. A reason for every line

A justified solution is a proof. Each line must be the line above it plus *one* named move:

- Addition / Subtraction / Multiplication / Division Property of Equality — the same operation on both sides.
- Distributive Property — rewrites *one* side; it is not a property of equality.
- Substitution Property — replaces a quantity by something already shown equal to it.

Example: Solve $4x - 9 = 15$, naming each reason.
Step 1. Every line needs one named move, so undo the operations one at a time rather than in a single jump.
Step 2. $4x - 9 = 15$ (Given); $4x = 24$ (Addition Property of Equality); $x = 6$ (Division Property of Equality).
$\Rightarrow \quad \boldsymbol{x = 6}$

Try it: Solve $3x + 8 = 29$ and name the reason on each line.

check: $\boldsymbol{x = 7}$ — Subtraction, then Division

2. From a segment picture to an equation

Segment Addition Postulate: if B is between A and C, then $AB + BC = AC$. Betweenness gives a **sum**.

Definition of midpoint: if M is the midpoint of $\overline{AB}$, then $AM = MB$, and each equals $\frac{1}{2}AB$. A midpoint gives an **equality between the parts** — and nothing else does. If no midpoint or congruence is stated, setting two parts equal does not follow.

Example: B is between A and C, $AB = x + 4$, $BC = 2x$, $AC = 19$. Find x.
Step 1. The only stated relationship is betweenness, so the postulate that applies is Segment Addition — add the parts, do not equate them.
Step 2. $(x + 4) + 2x = 19$, so $3x + 4 = 19$ and $3x = 15$.
$\Rightarrow \quad \boldsymbol{x = 5}$

Try it: B is between A and C, $AB = 3x - 2$, $BC = x + 6$, $AC = 28$. Find x.

check: $\boldsymbol{x = 6}$

3. From an angle relationship to an equation

Each relationship supplies a different number.

- Angle bisector: the two halves are *equal* to each other.
- Complementary: the two measures add to 90.
- Linear pair / supplementary: they add to 180.
- Vertical angles: equal to each other.
- Same-side interior angles, parallel lines: add to 180.

Name the relationship first — it is the reason, and it decides the equation.

Example: Two complementary angles measure $(2x)^\circ$ and $(3x+15)^\circ$. Find x.
Step 1. Complementary is the stated relationship, and complementary angles add to a right angle — so the equation is a sum, not an equality.
Step 2. $2x+(3x+15)=90$, so $5x+15=90$ and $5x=75$.
$\Rightarrow \quad \boldsymbol{x=15}$

Try it: Two supplementary angles measure $(4x+10)^\circ$ and $(2x+20)^\circ$. Find x.

check: $\boldsymbol{x = 25}$

4. Writing the two-column proof

Shape of the argument: start with the givens, end with the statement you were asked to prove, and let every line in between be licensed by the lines above it. Work *backwards* first — ask what would immediately give the conclusion, then what would give that.

Reasons come from four places only: Given, a definition, a postulate or theorem, or a property of equality. "It looks that way" is not one of them.

Example: Given M is the midpoint of $\overline{AB}$, prove $AM \cong \overline{MB}$.
Step 1. The conclusion is about the two halves, and the only given is the midpoint, so the proof is the definition of midpoint plus the link between equal measures and congruence.
Step 2. Each line below carries exactly one reason, ending on the statement that was to be proved.

Statements	Reasons
1. M is the midpoint of $\overline{AB}$	1. Given
2. $AM = MB$	2. Definition of midpoint
3. $\overline{AM} \cong \overline{MB}$	3. Definition of congruent segments

Try it: $\angle A$ and $\angle B$ are vertical angles. Prove $\angle A \cong \angle B$ in three lines — supply both columns.

check: Given; Vertical Angles Theorem; Definition of congruent angles